

MEDICAL HISTORY SUMMARY

Natural Death Investigation

Case Number:	ME-2024-003-NAT
Decedent:	CONFIDENTIAL
Date of Birth:	April 3, 1966
Age at Death:	58 years
Date of Death:	November 28, 2024
Compiled By:	Medical Examiner Investigator K. Thompson
Date Compiled:	November 29, 2024

SOURCES OF INFORMATION

Medical history compiled from: Riverside Medical Group (primary care physician Dr. James Reynolds, MD), St. Mary's Hospital emergency department records, cardiology consultation records, pharmacy records, family interviews (sister and ex-wife), and employer health records. All information obtained through proper legal authorization.

PRIMARY CARE PHYSICIAN RECORDS

Patient-Physician Relationship

The decedent established care with Dr. James Reynolds at Riverside Medical Group in March 2015. Medical records document approximately 14 office visits over the nine-year period. The most recent visit was March 2023, approximately 20 months prior to death. The patient had missed several scheduled follow-up appointments in 2024, including a missed appointment in September 2024.

Documented Medical Conditions

Essential Hypertension: Diagnosed in August 2016, approximately eight years before death. Initial blood pressure readings were 158/98 mmHg. The patient was counseled extensively regarding lifestyle modifications including diet, exercise, weight loss, and stress reduction. He was started on antihypertensive medication (lisinopril 10 mg daily) in October 2016 when lifestyle modifications proved insufficient. Over subsequent years, medication was adjusted multiple times due to

suboptimal blood pressure control. At the March 2023 visit, blood pressure was 168/102 mmHg despite prescribed medication (lisinopril 20 mg and hydrochlorothiazide 25 mg daily).

Pharmacy records indicate inconsistent prescription refills, suggesting poor medication adherence. The patient would often go several months without refilling prescriptions. When counseled about this, he reportedly acknowledged difficulty remembering to take medications and expressed concern about costs, though he had insurance coverage.

Dyslipidemia: First documented in 2017 with total cholesterol 278 mg/dL, LDL cholesterol 189 mg/dL, and triglycerides 245 mg/dL. The patient was counseled regarding dietary modifications and offered statin therapy. He declined medications initially, preferring to try diet and exercise. Follow-up testing in 2019 showed minimal improvement. Statin therapy (atorvastatin 20 mg) was prescribed in 2020 after repeated counseling. Subsequent testing showed some improvement but values remained above optimal levels, again suggesting inconsistent medication adherence.

Obesity: Documented throughout the medical record with BMI ranging from 29.8 to 32.1 (overweight to obese class I). Weight ranged from 92-98 kg. The patient received repeated counseling regarding weight loss through diet and exercise but had difficulty achieving sustained weight reduction. He reported limited time for exercise due to work demands.

Tobacco Use: The patient was a current smoker with approximately 30 pack-year history (smoking 1 pack per day for 30 years). He reportedly began smoking in his late teens. Multiple smoking cessation attempts were documented, including use of nicotine patches and prescription medications (varenicline), but he repeatedly relapsed. At his last visit, he expressed intention to quit but had not yet made a serious attempt. Dr. Reynolds extensively counseled regarding smoking's cardiovascular risks.

Prediabetes: Fasting glucose levels in 2022 and 2023 were in the prediabetic range (112-118 mg/dL). Hemoglobin A1c was 5.9% (prediabetic range 5.7-6.4%). The patient was counseled regarding progression risk to diabetes and need for lifestyle modifications. No diabetes medications were prescribed.

Laboratory Findings

Laboratory work from March 2023 (most recent): Complete Blood Count normal. Comprehensive Metabolic Panel: Glucose 115 mg/dL (prediabetic), creatinine 1.1 mg/dL (normal), estimated GFR 72 mL/min (mildly reduced), electrolytes normal. Lipid Panel: Total cholesterol 248 mg/dL (elevated), LDL 168 mg/dL (elevated), HDL 38 mg/dL (low - cardiovascular risk factor), triglycerides 210 mg/dL (elevated). These values indicated poorly controlled dyslipidemia despite statin therapy.

ECG performed in 2021 showed left ventricular hypertrophy (increased heart muscle thickness from chronic hypertension) and nonspecific ST-T wave changes. The patient was counseled that these findings indicated cardiac stress from hypertension and recommended strict blood pressure control. He was offered referral to cardiology, which he declined, stating he would focus on medication adherence and lifestyle changes.

Symptoms and Warning Signs

Review of medical records reveals several potentially significant symptoms that may have represented warning signs of cardiovascular disease: In November 2022, the patient reported

occasional chest discomfort with exertion that he attributed to 'indigestion' or muscle strain. He minimized these symptoms when questioned. In March 2023, he mentioned feeling more easily fatigued but attributed this to work stress and age. He denied classic angina symptoms when specifically asked. Dr. Reynolds documented concern about possible cardiac symptoms and recommended stress testing, which the patient agreed to but never scheduled.

In retrospect, these symptoms may have represented angina or other manifestations of underlying coronary disease. However, atypical presentation and patient minimization led to the symptoms not being fully evaluated. This is unfortunately common, as many cardiac patients have vague or atypical symptoms before a major event.

FAMILY AND SOCIAL HISTORY

Family History - Significant Risk Factors

Family history is highly significant for premature cardiovascular disease. The patient's father died of myocardial infarction at age 62. The patient's paternal grandfather also died of heart disease in his early 60s. One of the patient's brothers (age 54) has coronary artery disease and has undergone coronary artery bypass surgery. This strong family history of premature coronary disease represents a major genetic risk factor that was well-documented in the medical record and discussed with the patient on multiple occasions.

Employment and Lifestyle

The decedent worked as a supervisor at Thompson Manufacturing for 23 years. According to family and coworkers, his work was demanding with long hours and significant stress. He typically worked 50-60 hours per week. Family reported he prioritized work over health concerns and often canceled medical appointments due to work commitments. His diet reportedly consisted largely of fast food and processed foods, with limited fruits and vegetables. He had limited time for exercise due to work schedule.

Personal Life and Social Factors

The decedent was divorced approximately five years prior to death. He lived alone and had limited social support beyond coworkers and occasional contact with family. Ex-wife reported that he tended to minimize health concerns and avoid doctors. His sister stated he was reluctant to 'make a fuss' about symptoms and often said he would address health issues when work was less demanding. This pattern of delayed care-seeking and symptom minimization is unfortunately common and represents a barrier to preventive care.

CARDIOVASCULAR RISK ASSESSMENT

The decedent had multiple well-documented major cardiovascular risk factors that collectively created an extremely high-risk profile for coronary events:

- Age 58 (men >45 years have increased risk)

- Male sex (men have higher risk than premenopausal women)
- Hypertension, poorly controlled despite medications
- Dyslipidemia with elevated LDL and low HDL
- Current tobacco use (30 pack-year history)
- Obesity (BMI 32)
- Prediabetes
- Strong family history of premature coronary disease
- Sedentary lifestyle
- High-stress occupation
- Poor medication adherence
- Delayed medical follow-up

Using standard cardiovascular risk assessment tools, this patient would have been classified as high to very high risk for major cardiac events. The combination of multiple modifiable risk factors (hypertension, smoking, dyslipidemia, obesity) with non-modifiable risk factors (age, sex, family history) created substantial risk.

MISSED OPPORTUNITIES FOR INTERVENTION

This case represents a potentially preventable death with multiple missed opportunities for intervention. The patient had identified risk factors and received appropriate medical recommendations including medications for hypertension and dyslipidemia, smoking cessation support, and lifestyle modification counseling. However, suboptimal adherence to medications and recommendations, along with delayed follow-up care and failure to undergo recommended cardiac testing, resulted in disease progression.

The symptoms of exertional chest discomfort and fatigue in the year before death may have represented warning signs that, if fully evaluated with stress testing or other cardiac assessment, might have identified severe coronary disease requiring intervention such as angioplasty or bypass surgery. However, patient factors including symptom minimization, failure to schedule recommended testing, and competing work priorities prevented this evaluation.

SUMMARY AND RELEVANCE TO INVESTIGATION

The comprehensive medical history review reveals a 58-year-old male with multiple significant cardiovascular risk factors and documented cardiac abnormalities (ECG showing left ventricular hypertrophy) who experienced sudden cardiac death. The constellation of risk factors, family history,

and documented coronary risk factors provides complete context for understanding the autopsy findings of severe coronary atherosclerosis and acute myocardial infarction.

The medical history strongly supports natural death from cardiovascular disease and excludes alternative explanations. The patient's risk profile makes sudden cardiac death from myocardial infarction an expected potential outcome. The manner of death is unquestionably natural, resulting from progressive cardiovascular disease influenced by genetic and lifestyle factors.

Compiled by: K. Thompson, Medical Examiner Investigator

Date: November 29, 2024